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ABERDARE TECHNICAL COLLEGE

MODULE 3,TERM 3

UNIT:NETWORK DESIGN AND MANAGEMENT

CONTINUOUS ASSESSMENT TEST-CAT2

UNIT CODE: 0612 451 07A

ELEMENT:INSTALL COMPUTER NETWORK

Total marks: 50

Time 1hour 30 minutes

INSTRUCTIONS TO CANDIDATES

1. *This paper has **fifteen questions**. Answer **ALL** questions.*
2. *Write your name, registration number, class and date on the answer booklet.*
3. *The marks for each question or part are shown in brackets. **Each mark matches one point, item or command asked for.** For example, "state four" carries 4 marks.*
4. *In calculation questions, **write the formula first, then substitute, then give the answer**. An answer without a formula or working scores at most half of the marks for that part.*
5. *IP addresses and subnet masks must be written in **dotted-decimal notation** unless stated otherwise.*
6. *Router and switch commands must be written in **Cisco IOS syntax**.*
7. *Use the marks to decide how much to write and how long to spend (about 2 minutes per mark).*
8. *Calculators are allowed. Mobile phones are **not** allowed in the examination room.*
9. *Write neatly. Untidy work may lose marks.*

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QUESTIONS

1. List the **four** layers of the TCP/IP model. (4 marks)

2. State **two** functions of the Network layer of the OSI model. (2 marks)

3. Define each of the following terms:

(i) Classful IP addressing (1 mark)

(ii) Classless Inter-Domain Routing (CIDR) (1 mark)

4. State **two** differences between static and dynamic IP addressing. (2 marks)

5. State **two** differences between IPv4 and IPv6. (2 marks)

6. (a) State the default subnet mask of each of the following address classes: Class A, Class B and Class C. (3 marks)

(b) Identify the class of the address **192.168.5.7**. (1 mark)

7. (a) Write the formula used to calculate each of the following in CIDR:

(i) Number of host bits (1 mark)

(ii) Total number of host addresses in a network (1 mark)

(iii) Number of usable host addresses in a network (1 mark)

(b) A network has the address **192.168.10.0/26**. Showing your working, determine:

(i) The number of host bits (1 mark)

(ii) The total number of host addresses (1 mark)

(iii) The number of usable host addresses (1 mark)

8. Convert each of the following CIDR prefix lengths to a subnet mask in dotted-decimal notation:

(a) /24 (1 mark)

(b) /26 (1 mark)

(c) /28 (1 mark)

(d) /30 (1 mark)

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9. Convert each of the following subnet masks to a CIDR prefix length:

(a) 255.255.255.0 (1 mark)

(b) 255.255.255.224 (1 mark)

(c) 255.255.255.252 (1 mark)

10. A network must support **at least 50 usable host addresses**. Showing your working, determine:

(a) The number of host bits needed. (1 mark)

(b) The CIDR prefix length. (1 mark)

(c) The subnet mask in dotted-decimal notation. (1 mark)

(d) The number of usable host addresses the network provides. (1 mark)

11. State **two** advantages of CIDR over classful IP addressing. (2 marks)

12. (a) Name the protocol used to assign IP addresses to devices dynamically. (1 mark)

(b) State **one** reason why a router interface should use a static IP address. (1 mark)

13. Write the Cisco IOS command for each of the following tasks on a router:

(a) Set the hostname to **R1**. (1 mark)

(b) Set an encrypted privileged EXEC password **kcse2026**. (1 mark)

(c) Enter configuration mode for interface **GigabitEthernet0/0**. (1 mark)

(d) Assign the IP address **192.168.10.1** with the mask for a /26 network. (1 mark)

(e) Activate the interface. (1 mark)

(f) Save the running configuration to NVRAM. (1 mark)

14. Write the **three** commands used to configure the management interface (VLAN 1) of a switch with the IP address **192.168.10.2** and the mask for a /26 network, and make the interface active. (3 marks)

15. Router R1 connects to the network **10.0.0.0/8** and to the LAN **192.168.10.0/24**. Write the **four** commands needed to configure **RIP version 2** on R1 so that both networks are advertised. (4 marks)

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1 Application Layer
2 Transport layer
3 Internet Layer
4 Network Access Layer

2) IP addressing
Routing packets between networks

3 Classful IP addressing: An IP addressing system that divides addresses into Classes A, B, C, D and E.

ii) Classless Inter-Domain Routing (CIDR): An IP addressing method that uses variable-length network prefixes

4 Static IP addresses are assigned manually, while dynamic IP addresses are assigned automatically.
2 Static IP addresses remain fixed, while dynamic IP addresses may change.

5) IPv4 uses 32-bit addresses, while IPv6 uses 128-bit addresses.
IPv4 is written in dotted-decimal notation, while IPv6 uses Hexadecimal notation

6) Class A - 255.0.0.0
Class B: 255.255.0.0
Class C - 255.255.255.0

b) Class C.

7) i) Host bits = 32 - Prefix Length
ii) 2^n, where n = number of host bits
iii) $2^n - 2$

b) $32 - 26 = 6$ bits
ii) $2^6 = 64$ addresses

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iii) $64-2=62$ addresses ✓ 0/1

6

9

10) $2^6-2=62$ usable addresses ✓
$=6$ bits ✓ 0/1

b $32-6=126$ ✓ 0/1

c 255.255.255.192 ✓ 0/1

d $2^6-2=62$ addresses ✓ 0/1

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11 Reduces wastage of IP addresses ✓ 8
Allows more efficient routing through route aggregation ✓ 8

12 DHCP ✓ 8
b) To ensure the IP address remains fixed ✓ 8

13 Set the hostname to R1
hostname R1 ✓ 8

b) Enable secret ksc_2025 ✓

c) Interface Gigabit Ethernet0/0 ✓ 8

d) ip address 192.168.10.1 255.255.255.192 ✓ 8

e) no shutdown ✓ 8

f) Copy running-config startup-config ✓ 8

14. interface vlan1 ✓ 8
ip address 192.168.10.2 255.255.255.192 ✓ 8
no shutdown ✓ 8

15 router rip ✓ 8
Version 2 ✓ 8
Network 10.0.0.0 ✓ 8
Network 192.168.10.0 ✓ 8

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